

Course Overview

Python for AI-Driven Automation and Business Data Science

Course materials

Contents

- 1 The big picture
- 2 Structure
- 3 Learning paths
- 4 How to study
- 5 What you'll build
- 6 Getting started

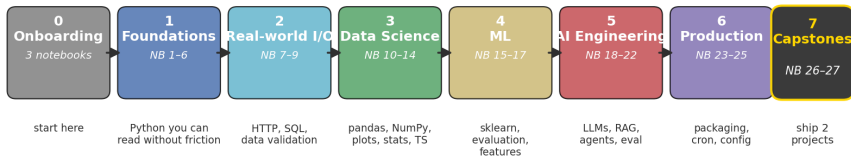
Outline

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The course in one diagram

Python for AI-Driven Automation and Business Data Science

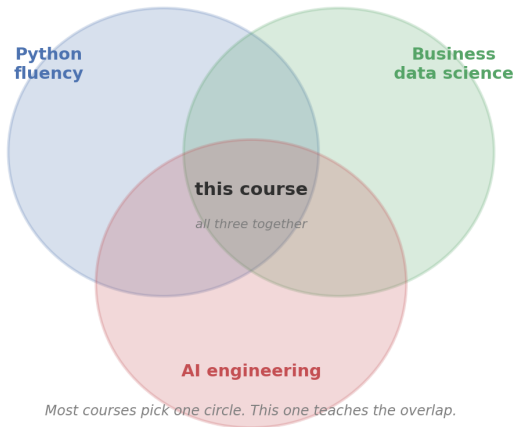
27 notebooks · 7 modules · 25–35 hours of focused study



— runs entirely offline · every exercise has a worked solution · all 27 notebooks tested in CI —

Foundations of every page: read prose → run code → tweak → predict → re-run

Where this course sits



Three skills, taught together

1. Python fluency

Read, write, and modify clean Python without friction. Functions, modules, error handling, defensive parsing.

2. Working with data professionally

pandas, NumPy, matplotlib, scikit-learn, statistics, time-series forecasting. Real datasets, not toy ones.

3. AI-driven automation

Calling LLMs as functions. Structured output. Retrieval-augmented generation. Tools and agents. Document processing. *Plus* the engineering wrapper around all of it.

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Seven modules, twenty-six notebooks

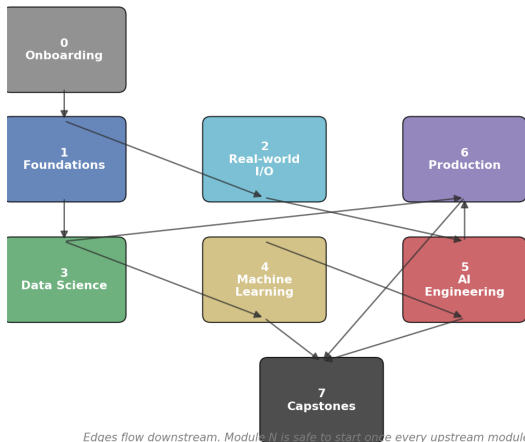
#	Module	Notebooks
0	Onboarding	1 — master onboarding
1	Foundations	NB 1–5 — Python you can read without friction
2	Real-world I/O	NB 7–8 — HTTP, SQL (<i>data validation reserved</i>)
3	Data science	NB 10–14 — pandas, NumPy, plots, stats, time series
4	Machine learning	NB 15–17 — sklearn, model evaluation, features
5	AI engineering	NB 18–22 — LLM patterns, RAG, agents, evaluation
6	Production	NB 23–24 — packaging, scheduling (<i>config reserved</i>)
7	Capstones	NB 26–27 — analytics + AI assistant

Every module folder has its own README

Goals, prerequisites, what you'll build, the notebooks in order. Read it before you open the first notebook of the module.

Module dependencies — what comes before what

Module dependency graph — what depends on what

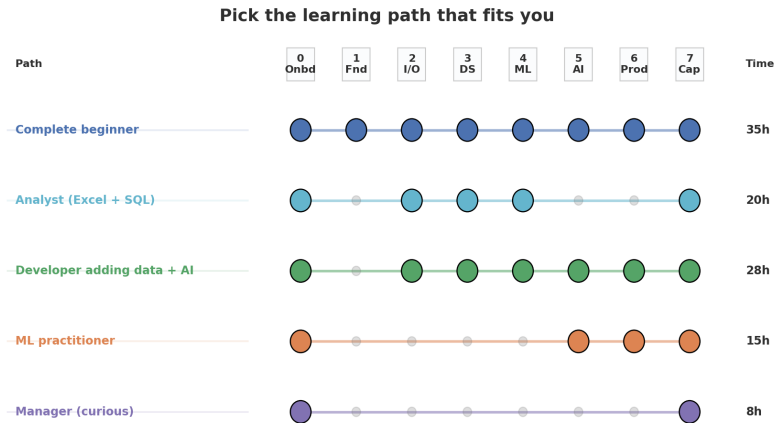


Edges flow downstream. Module N is safe to start once every upstream module is comfortable.

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Pick the path that fits you



Each dot is one module. Solid line = modules you'd touch on that path.

When to pick which path

Complete beginner

Never written code before. Take all seven modules in order. Time-budget: **35 hours**, split into 6 weeks of ~ 6 h each.

Analyst who knows Excel and SQL

Skim Module 1 (Python basics) and Modules 2 (SQL is already familiar). Focus on Modules 3 (DS), 4 (ML), 7 (capstone). Time-budget: **20 hours**.

ML practitioner adding AI engineering

Skim Module 4. Focus on Modules 5 (AI), 6 (Production), 7 (Capstone). Time-budget: **15 hours**.

Six weeks at six hours each

A self-paced 6-week schedule

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Onboarding + Foundations	Real-world I/O	Data Science (pandas, statsmodels)	Time series, ML basics	AI Engineering	Production + capstone
NB 0-6	NB 7-9	NB 10-13	NB 14-17	NB 18-22	NB 23-27
8h focus	5h focus	6h focus	6h focus	7h focus	6h focus

≈ 6 hours of focused study per week · more time → faster, less time → split a week in two

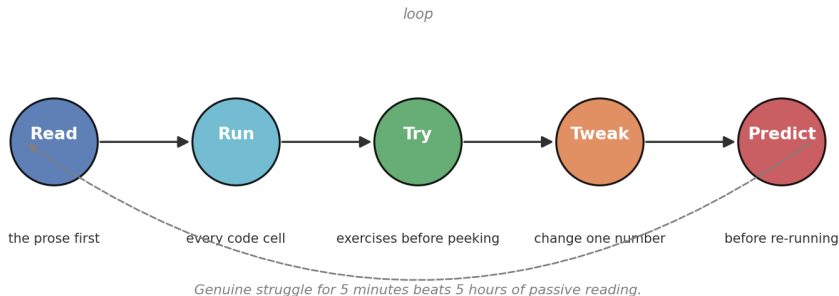
Got more time? Move faster. Got less? Split one week into two.

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The five-step study loop

The 5-step study loop — apply to every notebook



Apply this loop to *every* notebook. Genuine struggle beats passive reading.

Six visual markers you'll see throughout

Marker	Meaning
(Tip)	A small “stop and notice” callout.
(Intuition)	The mental model behind the syntax.
(Pitfall)	A bug or anti-pattern that bites if you're not warned.
(Exercises)	Hands-on tasks. Try <i>before</i> peeking at the solution.
(Bonus)	One larger applied task per notebook.
(Debug me)	A cell that <i>intentionally</i> contains a bug for you to find.
(Next step)	Always points to the next notebook in your path.

Once you internalise the markers they become road signs, not decoration.

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Six small artefacts you ship along the way

Six small artefacts you'll ship by the end

KPI snapshot

An AI-bot KPI report
in ~30 lines of Python

Live API ETL

Daily weather pulled
from Open-Meteo

SQL report

Channel summary written
in SQL + pandas

Forecast

3-month forecast
with Holt-Winters

Inbox triage

LLM that classifies +
summarises feedback

Scheduled job

Weekly report task
with retries + alerts

Plus two big capstones at the end

Capstone A — AI Support-Bot Analytics (NB 26)

Five channels $\times$ twelve months of operational data $\rightarrow$ 2×2 executive dashboard $\rightarrow$ regression demonstrating Simpson's paradox $\rightarrow$ five-bullet executive summary.

Capstone B — AI Customer-Feedback Assistant (NB 27)

Ingest free-form customer feedback $\rightarrow$ classify + extract via the MockLLM $\rightarrow$ ground answers in a knowledge base via RAG $\rightarrow$ orchestrate as a scheduled job with alerts. End-to-end AI feature.

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LLM providers — local and hosted, four options

Provider	Class	Best for
OpenAI	<code>OpenAILLM("gpt-4o-mini")</code>	Reliable default
Anthropic	<code>AnthropicLLM("claude-haiku")</code>	Long context, careful tone
Google	<code>GoogleLLM("gemini-2.0-flash")</code>	Cheapest at high volume
Ollama	<code>OllamaLLM("llama3.2:3b")</code>	Local – no internet, no key

Default = the offline MockLLM

Every AI notebook runs offline by default. Swapping in a real provider is a *one-line* change thanks to the unified interface in `refined_course/llm_providers.py`.

See the dedicated guide

Notebook `05_ai_engineering/A1_llm_providers_guide.ipynb` has the full setup walk-through, model recommendations, and cost estimates.

What you need to install

For Google Colab

Nothing. All required libraries are pre-installed. Open the notebook, hit Shift+Enter.

For local Jupyter

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
jupyter lab
```

Notebook 0 includes an environment-check cell you run to confirm everything works before you move on.

Now → open Notebook 0

One file to open. The rest unfolds from there.

`00_onboarding/00_master_onboarding.ipynb`

Three things that notebook does for you

- 1 Confirms your Python environment is correctly set up.
- 2 Explains the five-step study loop you'll use throughout.
- 3 Helps you pick the right learning path from the five options.

One last reminder before you start

Five-step study loop, in your pocket

Read the prose first → **Run** every code cell → **Try** the exercises before peeking → **Tweak** one number → **Predict** the new output before re-running.

When to use which marker

(Tip) information; (Intuition) mental model; (Pitfall) avoid this; (Exercise) hands-on practice; (Bonus) bigger applied task; (Debug me) intentional bug — find it.

When you're stuck for 15 minutes

Copy the full error into a search engine. 95% of beginner errors are well-documented. The course is designed to teach you to debug, not to spare you the experience.

Welcome.

26 notebooks await you.

Open Module 0 next.